

A Study on High-Speed and High-Precision Tracking Control of Large-Scale Stage Using Perfect Tracking Control Method Based on Multirate Feedforward Control

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Abstract—In hard-disk-drive control, the perfect tracking control (PTC) method has already obtained high performance. Therefore, the first author and his group study the PTC scheme for improving the performance of large-scale stages. This paper presents an application of the PTC scheme to the tracking control of two different stages. The results from both simulations and experiments show that the PTC method outperforms the conventional rigid-body-mode-based feedforward method.

Index Terms—High precision, high speed, motion control systems, positioning table.

I. INTRODUCTION

STAGES are widely used as positioning devices for manufacturing and measurement processes in a variety of industrial machinery [1], [2]. Such stages have been consistently required to achieve high speed and high precision to satisfy the need for improving production efficiencies. For stages incorporated in semiconductor manufacturing equipment, these two requirements are extremely important. This has led to tireless and intense efforts on the research and development of advanced control technologies that aim at achieving high speed and high precision.

Under these circumstances, the iterative learning control (ILC) proposed over ten years ago continues to be studied vigorously [3]–[7]. In most cases, however, ILC assumes that a stage is highly rigid with no mechanical resonances in the frequency range below several hundreds of hertz. When this prerequisite is satisfied, the use of ILC enables a high-gain high-bandwidth feedback control, thereby achieving high speed and high precision. However, with an elastic stage with many mechanical resonances in the lower frequency range, a high-gain high-bandwidth feedback control cannot be implemented. This is because vibrations are excited and it is difficult to ensure a stable closed loop. Consequently, it is quite difficult to satisfy

the requirements of high speed and high precision with a control method that primarily uses ILC.

One of the applications for an elastic stage with many mechanical resonances in the lower frequency range is industrial machinery for processing or measuring large-scale parts. A stage that is incorporated in such machinery faces an additional challenge of having to increase its size to help improve production efficiencies on top of achieving high speed and high precision. The larger the stage becomes for better production efficiency, the greater the number of mechanical resonances tends to become at lower frequencies. A large-scale stage, which is the intended application of the method proposed in this paper, has a number of mechanical vibration modes at frequencies of several tens to several hundreds of hertz. How well the requirements of high speed and high precision can be achieved without exciting these vibration modes is the paramount control challenge for such large-scale stage.

As a technique for controlling an object with low-frequency vibration modes as in a large-scale stage, a vibration-suppression control method based on the two-mass model has been proposed [8], [9]. This method attempts to realize a vibration-suppression control only with a feedback control, and it works well in controlling a simple structure like the plants presented in the papers in which higher order vibration modes, except the primary-resonance mode, are found to be negligible. Trying to realize high gain and high bandwidth using this method in an elastic stage with many mechanical resonances, however, would be difficult in terms of ensuring a stable closed loop as in the case with ILC.

Furthermore, a large-scale stage, which is the intended application of this paper, has multiple axes, resulting in a complicated structure. This adds yet another factor in increasing mechanical resonances. Mechanical resonances resulting from multiaxiality are such that a closed loop can be stabilized for each axis. However, vibrations are often excited by a mutual coupling loop because of a mechanical structure and a controller that are coupled together as the loop gain is increased. Considering these points, it is difficult for feedback-control-based vibration suppression and the requirements for high speed and high precision to be satisfied simultaneously.

For the reasons thus far explained, a large-scale stage requires a control method that can satisfy the requirements of high speed, high precision, and larger size without having to

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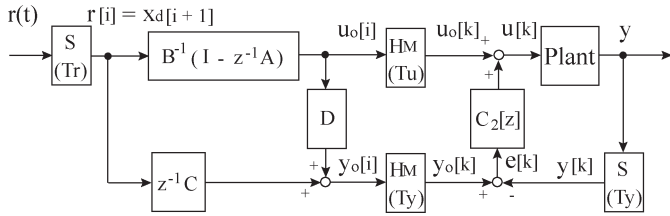


Fig. 1. PTC system.

depend on a high-gain high-feedback control. As a control method that satisfies this demand, the authors propose a 2-DOF control method that enables an accurate feedforward control to a maximum extent. For the method proposed in this paper, how accurately an inverse system for a plant can be reflected in a feedforward control is the key to achieving the performance required. If a controller is assumed to be configured with a digital computing unit, this method has to be designed based on a discrete-time system.

Tomizuka *et al.* [10], [11] proposed the zero-phase-error tracking controller which compensates both phase and gain errors caused by uncancellable unstable zeros of a discrete-time plant.

Whereas Fujimoto, a coauthor of this paper, proposed a perfect tracking control (PTC) based on multirate feedforward control to eliminate the problem of unstable zeros [12]. PTC theoretically guarantees that a feedforward control based on an accurate inverse system for a nominal model can be realized and that perfect tracking for every sampling period can be achieved as long as there is no unstable zero in the nominal model defined by a continuous-time system. In recent years, he has also proposed vibration-suppression PTC that takes into account resonance modes of a control object, expanding the scope of application of the PTC method [13], [14]. These practical control methods were first successfully applied to a hard-disk-drive (HDD) control.

A precision stage shares commonalities with HDD in that they are both driven by a voice coil motor or linear motor and required to achieve control specifications of high speed and high precision. Therefore, the authors attempt to apply the PTC method to a large-scale high-precision stage to improve the control performance. This paper describes the results of the application of the PTC method to two different large-scale stages.

II. PTC

A. Basic Structure of PTC [12]

First, we explain the basic structure of PTC. PTC has a structure of a 2-DOF control system, as shown in Fig. 1. This system has two samplers for the reference signal $r(t)$ and the output $y(t)$ and two holders for the input $u_o(t)$ and nominal output $y_o(t)$. Therefore, there exist sampling periods T_r , T_y , and T_u which represent the periods of $r(t)$, $y(t)$, and $u(t)$, respectively. PTC applies the multirate feedforward control in which the control input $u(t)$ is changed n times during one sampling period T_r of reference input $r(t)$, as shown in Fig. 2.

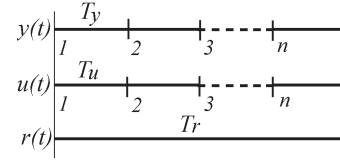


Fig. 2. Multirate sampling period.

Here, n is the plant order. H_M in Fig. 1 is the multirate holder which outputs the input $u[i] = [u_1[k] \cdots u_n[k]]^T$ (generated by the long sampling period T_r) on the short sampling period T_u .

Here, the plant discrete-time state-space matrices A , B , C , and D at the long sampling period T_r can be derived as (2) from the discrete-time plant model at the short sampling period T_u

$$x[k+1] = A_s x[k] + b_s u[k]$$

$$y[k] = c_s x[k] \quad (1)$$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A_s^n & A_s^{n-1}b_s & \cdots & A_s b_s & b_s \\ c_s & 0 & \cdots & 0 & 0 \\ c_s A_s & c_s b_s & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ c_s A_s^{n-1} & c_s A_s^{n-1} b_s & \cdots & c_s b_s & 0 \end{bmatrix}. \quad (2)$$

Since the matrix B of (2) is nonsingular in the case of a controllable plant, PTC can be designed as

$$\begin{aligned} u_o[i] &= B^{-1}(I - z^{-1}A)x_d[i+1] \\ &= \begin{bmatrix} 0 & I \\ -B^{-1}A & B^{-1} \end{bmatrix} x_d[i+1] \end{aligned} \quad (3)$$

$$y_o[i] = z^{-1}C x_d[i+1] + D u_o[i]. \quad (4)$$

Equation (3) is the stable inverse system of a plant with input from the previewed desired trajectories. Therefore, perfect tracking is assured at the sampling period T_r .

The feedback control $C_2[z]$ suppresses the error between the output $y[k]$ and the nominal output $y_o[k]$ caused by disturbance or feedforward control imperfectness due to plant model error.

B. Vibration-Suppression PTC [13]

In order to suppress the excitation of the resonance modes of the plant, the vibration-suppression PTC is introduced [13], in which the nominal model is defined as rigid-body and primary-resonance modes. Moreover, the high-order vibration-suppression PTC was proposed in [14] to suppress the second and higher resonance modes. Generally, the plant model which consists of rigid-body and resonance modes is defined as

$$P(s) = \frac{b_{n-1}s^{n-1} + \cdots + b_1s + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_1s + a_0}. \quad (5)$$

The controllable canonical form of (5) with state variables $\mathbf{x} = [z, z^{(1)}, z^{(2)}, \dots, z^{(n-1)}]^T$ is represented by

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \mathbf{A}_c \mathbf{x}(t) + \mathbf{b}_c u(t) \\ y(t) &= \mathbf{c}_c \mathbf{x}(t) \end{aligned} \quad (6)$$

$$\mathbf{A}_c = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & \ddots & 0 \\ 0 & \cdots & \cdots & 0 & 1 \\ -a_0 & -a_1 & \cdots & -a_{n-2} & -a_{n-1} \end{bmatrix} \quad \mathbf{b}_c = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ b_0 \end{bmatrix} \quad (7)$$

$$\mathbf{c}_c = \left[1 \quad \frac{b_1}{b_0} \quad \frac{b_2}{b_0} \quad \cdots \quad \frac{b_{n-1}}{b_0} \right]. \quad (8)$$

Equations (3) and (4) of vibration-suppression PTC can be designed in the same way as in Section II-A.

Here, the details of the input method of target trajectories in the case of vibration-suppression PTC are described as follows.

As we define $x_1(t) = z(t)$, the target trajectory $\mathbf{x}_d(t)$ is represented by

$$\mathbf{x}_d(t) = \left[z_d(t), z_d^{(1)}(t), z_d^{(2)}(t), z_d^{(3)}(t), \dots, z_d^{(n-1)}(t) \right]^T \quad (9)$$

where $z_d^{(n)}(t)$ is the n th-order differentiation by definition of the controllable canonical form. However, $z_d(t)$ is not the real target trajectory because the values of $b_1/b_0, b_2/b_0, \dots, b_{n-1}/b_0$ are not zeros in (8). Therefore, $z_d(t)$ is called “virtual position.” If it is directly applied to the target trajectory, perfect tracking cannot be achieved in tracking motion.

In this case, a “target trajectory filter” is inserted to generate the target trajectory $z_d(t)$ from the real position trajectory $y_d(t)$ [13]. In this manner, PTC and vibration-suppression control can be achieved in tracking motion.

III. STRUCTURES AND MODELING OF STAGE

A. Structures of Stage

The large-scale high-precision stage with degrees of freedom in the XY- or XY θ -axes can be classified into two forms in terms of the difference in the Z-axis air guide.

One is the form for realizing the aforementioned degree of freedom at the single-layer Z-axis air guide, as shown in Fig. 3(a). It is called planar stage. Another is the form with an individual multilayer Z-axis air guide for each degree of freedom, as shown in Fig. 3(b). It is called gantry stage.

B. Modeling of Planar Stage

Even if a planar stage is not constrained in the XY-direction, it can be modeled as a single-input–single-output (SISO) system. That is because it is clear that its motion almost follows the equation of motion for each axis. At this time, the transfer function of the plant model is represented by

$$P_1(s) = \frac{p}{f} = \frac{y}{u} = \frac{b_0}{a_2 s^2 + a_1 s}, \quad a_2 = M; \quad a_1 = C; \quad b_0 = 1 \quad (10)$$

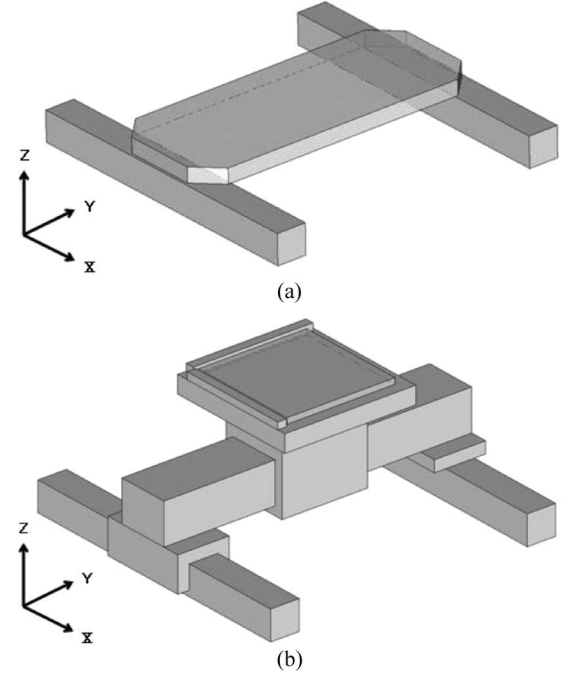


Fig. 3. Structures of stages. (a) Planar stage. (b) Gantry stage.

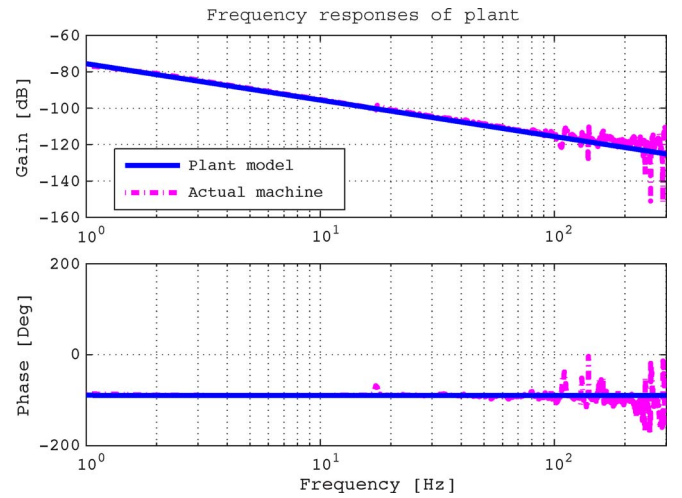


Fig. 4. Frequency responses of a plant.

where p is the stage x or y position, f is the associated control force, M is the mass of stage, and C is the viscosity of stage.

Fig. 4 shows the frequency responses of the plant model and actual machine in the X-direction. From this figure, both responses are almost identical.

C. Modeling of Gantry Stage

The gantry stage can be modeled as a mechanically separated SISO system since it has an individual multilayer Z-axis air guide for each degree of freedom. However, due to the multilayer configuration, it is generally difficult to adjust the Z position of the actuator to match the Z position of the stage center of gravity. For this reason, if that structure is simplified, it can be shown as Fig. 5. At this time, if θ is very small,

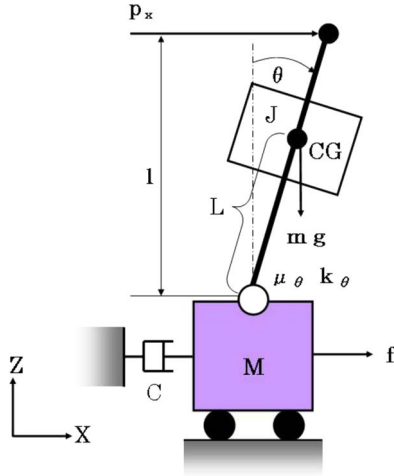


Fig. 5. Model of a plant.

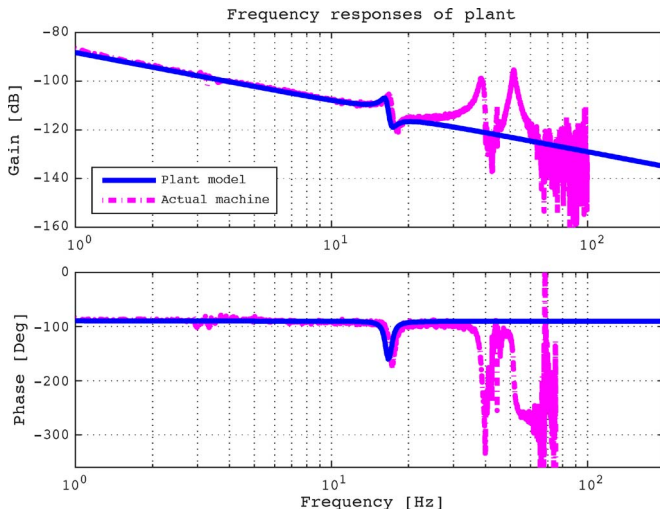


Fig. 6. Frequency responses of a plant.

the transfer function of the plant model is represented by (11). Here, this system is a minimum-phase system

$$P_2(s) = \frac{p_x}{f} = \frac{y}{u} = \frac{b_2 s^2 + b_1 s + b_0}{a_4 s^4 + a_3 s^3 + a_2 s^2 + a_1 s}, \quad (11)$$

$$a_4 = MmL^2 + MJ + mJ$$

$$a_3 = M\mu_\theta + m\mu_\theta + (mL^2 + J)C$$

$$a_2 = Mk_\theta + mk_\theta - MmgL - m^2gL + \mu_\theta C$$

$$a_1 = (k_\theta - mgL)C$$

$$b_2 = mL^2 + J - mL$$

$$b_1 = \mu_\theta$$

$$b_0 = k_\theta - mgL$$

where each coefficient is shown in Fig. 5.

Fig. 6 shows the frequency responses of the plant model and actual machine, and they are almost identical up to the primary-resonance mode.

IV. DESIGN OF CONTROLLER

As explained in Section II, since PTC has the structure of a 2-DOF control system, the feedforward and feedback controllers can be designed independently. In this section, the design methods for each controller are explained.

A. Design of the Proposed Feedforward Controller

The controllable canonical form of the plant model is necessary for designing the feedforward controllers based on PTC. In the planar stage with (10), its equation is represented by

$$\dot{\mathbf{x}}(t) = \mathbf{A}_c \mathbf{x}(t) + \mathbf{b}_c u(t)$$

$$y(t) = \mathbf{c}_c \mathbf{x}(t) \quad (12)$$

$$\mathbf{x}(t) = [z(t), z^{(1)}(t)]^T$$

$$\left[\begin{array}{c|c} \mathbf{A}_c & \mathbf{b}_c \\ \hline -\mathbf{c}_c & 0 \end{array} \right] = \left[\begin{array}{cc|cc} 0 & 1 & 0 & 0 \\ 0 & -\frac{a_1}{a_2} & \frac{b_0}{a_2} & \frac{b_1}{a_2} \\ \hline 1 & 0 & 0 & 0 \end{array} \right]. \quad (13)$$

In the gantry stage with (11), its equation is represented by

$$\dot{\mathbf{x}}(t) = \mathbf{A}_c \mathbf{x}(t) + \mathbf{b}_c u(t)$$

$$y(t) = \mathbf{c}_c \mathbf{x}(t) \quad (14)$$

$$\mathbf{x}(t) = [z(t), z^{(1)}(t), z^{(2)}(t), z^{(3)}(t)]^T$$

$$\left[\begin{array}{c|c} \mathbf{A}_c & \mathbf{b}_c \\ \hline -\mathbf{c}_c & 0 \end{array} \right] = \left[\begin{array}{cccc|cc} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & -\frac{a_1}{a_4} & -\frac{a_2}{a_4} & -\frac{a_3}{a_4} & \frac{b_0}{a_4} & \frac{b_1}{a_4} \\ \hline 1 & \frac{b_1}{b_0} & \frac{b_2}{b_0} & 0 & 0 & 0 \end{array} \right]. \quad (15)$$

Here, as the equation of the plant model discretized by the short sampling period T_u is (1), the feedforward controllers based on PTC can be obtained from (3).

However, the sampling periods of these stage controllers are different. The planar stage's sampling period is $T_u = T_y = T_r/2$, and the one for the gantry stage is $T_u = T_y = T_r/4$.

Moreover, the target trajectories of these stage controllers are different, too. In the planar stage, since $y(t) = x_1(t) = z(t)$ from (13), the target trajectory $z_d(t)$ can be used directly. In the gantry stage defined as rigid-body and primary-resonance modes, however, it cannot be given directly as a reference because $y(t) \neq x_1(t) = z(t)$ from (15) and $x_1(t)$ is a virtual position, as explained in Section II-B. Therefore, using the target trajectory filter of (16) [15], it is inserted to generate the target trajectory $z_d(t)$ from the real position trajectory $y_d(t)$. Then, the zeros of the plant model are canceled, and PTC can be realized

$$R(s) = \frac{b_0}{b_2 s^2 + b_1 s + b_0}. \quad (16)$$

B. Design of Feedback Controller

For both stages, the proportional–proportional and integral control system which has a velocity loop inside a position loop is used as the feedback controller. In the case of using an approximated simple plant model, feedback controllers are apt to be designed conservatively. Therefore, those parameters

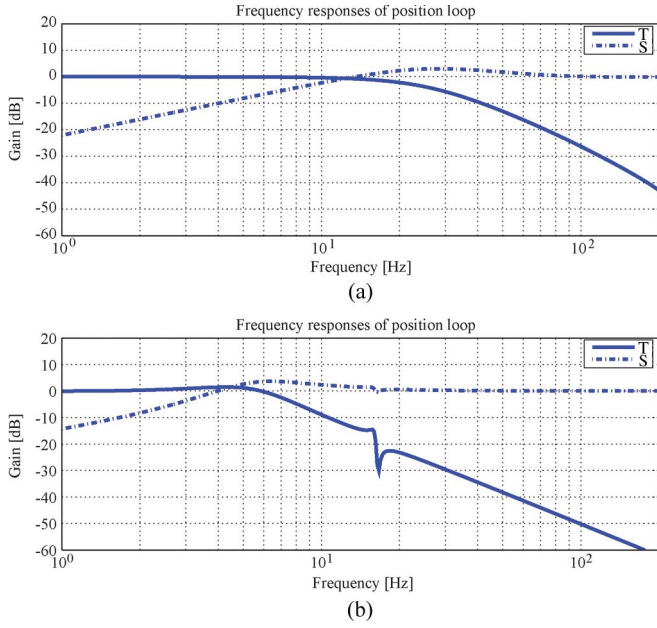


Fig. 7. Frequency responses of a position loop. (a) Planar stage. (b) Gantry stage.

were designed through a trial-and-error process in order to avoid a conservative design. In the case of a large-scale stage having many vibration modes, the stability issue induced by higher order resonances and the influence of vibration excitation should be verified. Therefore, when dealing with these problems, a more practical and better performance controller can be realized when it is designed through a trial-and-error process purely based on a real drive condition, as opposed to analytically designing it by using a simple plant model with a definition of modeling errors.

As a result, parameters are set such that none of the peak gains of higher order resonances that were ignored exceeds -10 dB in the open-loop characteristic. Fig. 7 shows the frequency responses of the position loop after the tuning. Fig. 7(a) shows those of a planar stage. Fig. 7(b) shows those of a gantry stage.

C. Design of Conventional Feedforward Controller

In the simulations and experiments in the following section, the proposed method is compared with a conventional method. The conventional feedforward controller used in the comparison will be explained next.

A conventional feedforward controller assumes that the plant model is equal to a rigid model. Moreover, using the differential trajectory of the target trajectory $r(t)$, the output of controller $u_{FF}(t)$ is calculated as

$$u_{FF}(t) = h_2 \ddot{r}(t) + h_1 \dot{r}(t) \quad (17)$$

where the coefficients of h_2 and h_1 are theoretically equal to the mass and viscosity of a rigid model. However, in reality, they are determined through a trial-and-error process. As shown in Fig. 8, this controller is combined with the previous feedback controller. As a result, the conventional control system is single-rate control.

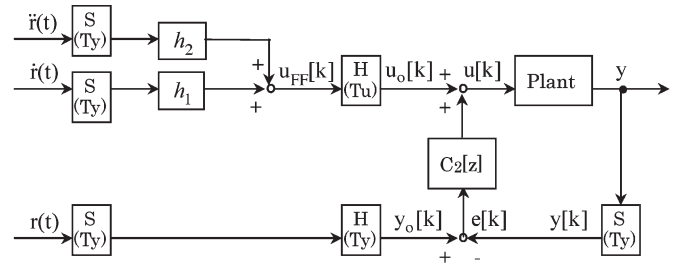


Fig. 8. Conventional control system.

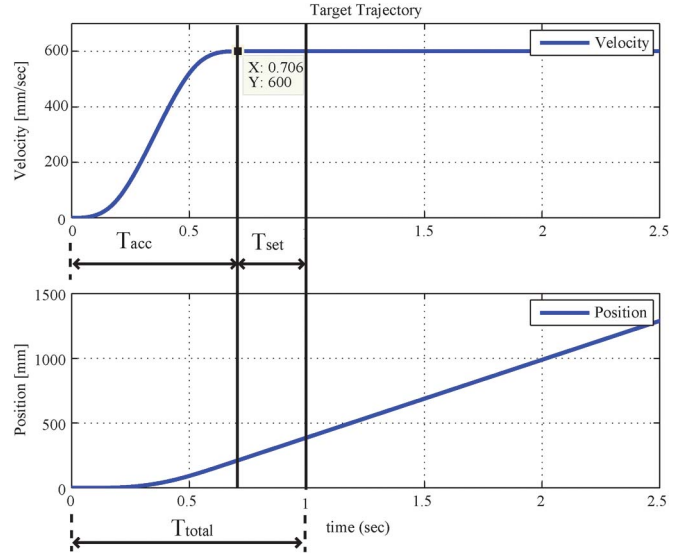


Fig. 9. Target trajectory.

V. SIMULATION

For the plant model, the simulations of the conventional and proposed methods are compared. Simulations were performed using Matlab/Simulink.

A. Definition of Specification and Control Performance

Each sampling period of the controller is $T_y = T_u = 1/3$ ms, which is the same with that of the actual controller.

For the proposed feedforward controller, the target position trajectory is given. In addition, the target velocity, acceleration, and jerk trajectories are given by differentiating the target position trajectory. Fig. 9 shows the target position and velocity trajectories. Here, t_{acc} is the time between the beginning of acceleration and the point where the acceleration becomes zero. In other words, it is the time taken for the velocity to become constant. Then, the constant velocity is defined as v_{const} . After the time of the end of t_{acc} , v_{const} does not change. The settling time t_{set} is the time it takes for a tracking error to be within a desired range s_{range} . The total time t_{total} is represented by

$$t_{total} = t_{acc} + t_{set}. \quad (18)$$

t_{set} and t_{total} are very important specifications for the high-speed and high-precision tracking stage because t_{set} and t_{total} should be shortened as much as possible to improve throughput.

TABLE I

CHARACTERISTIC AND SIMULATION RESULTS OF THE PLANAR STAGE.
(a) CHARACTERISTIC OF TARGET TRAJECTORY AND SETTLING RANGE.
(b) SIMULATION RESULTS OF t_{set} AND t_{total}

(a)		
v_{const} [mm/s]	t_{acc} [sec]	s_{range} [count]
100	0.5	± 200

(b)		
Method	t_{set} [sec]	t_{total} [sec]
Conventional	0.3	0.8
Proposed	0.0	0.5

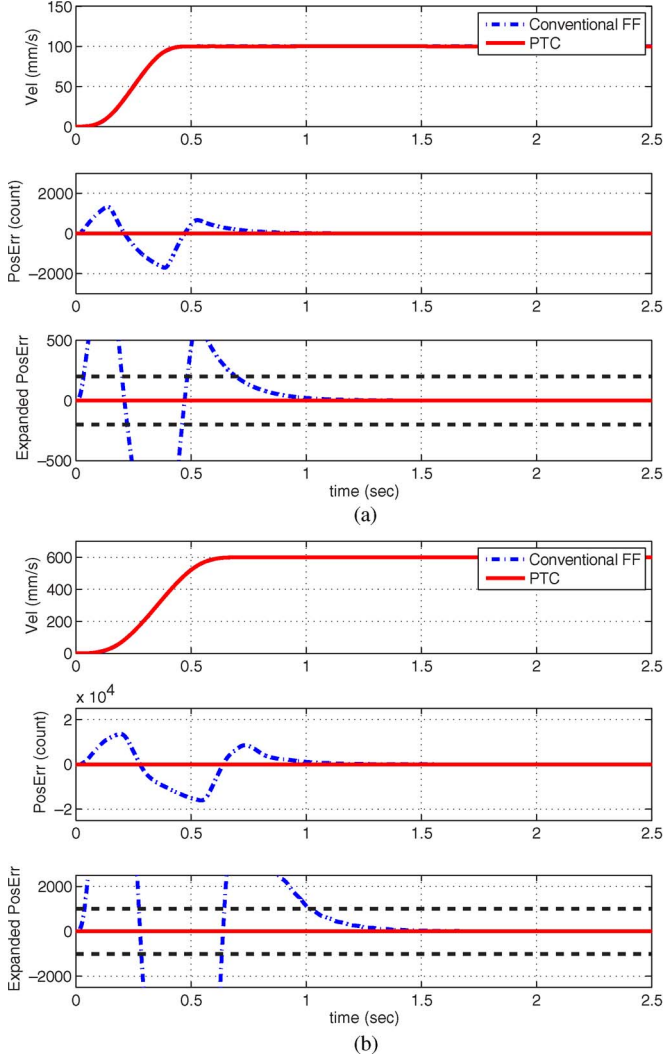


Fig. 10. Simulations of time response. (a) Planar stage. (b) Gantry stage.

B. Simulation of Planar Stage

The specifications of the target trajectory and the settling range s_{range} are shown in Table I(a). This value of the s_{range} indicates that the precision is on the order of submicrometers because the resolution of the position sensor is on the order of nanometers. The simulation results of t_{set} and t_{total} shown in Fig. 10(a) are summarized in Table I(b).

As shown in Table I(b), $t_{set} = 0$ is achieved in the proposed method.

TABLE II

CHARACTERISTIC AND SIMULATION RESULTS OF THE GANTRY STAGE.
(a) CHARACTERISTIC OF TARGET TRAJECTORY AND SETTLING RANGE.
(b) SIMULATION RESULTS OF t_{set} AND t_{total}

(a)		
v_{const} [mm/s]	t_{acc} [sec]	s_{range} [count]
600	0.706	± 1000

(b)		
Method	t_{set} [sec]	t_{total} [sec]
Conventional	0.3	1.0
Proposed	0.0	0.7

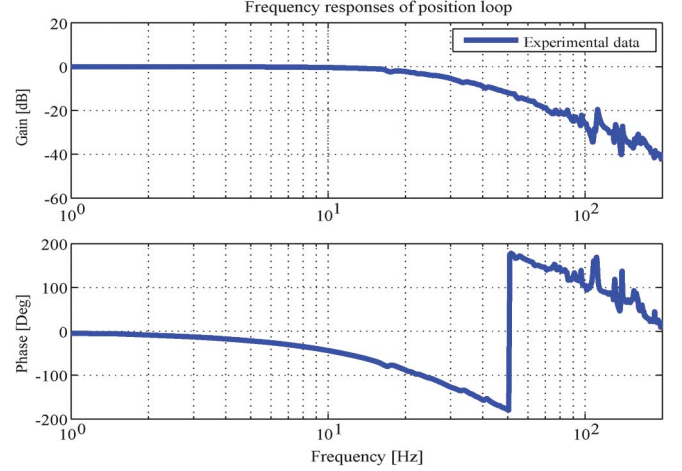


Fig. 11. Frequency responses of a position loop (planar stage).

C. Simulation of Gantry Stage

The specifications of the target trajectory and the settling range s_{range} are shown in Table II(a). Furthermore, the simulation results of t_{set} and t_{total} shown in Fig. 10(b) are summarized in Table II(b).

The proposed method has the same result as the planar stage. On the other hand, the conventional method becomes worse than the planar stage because the bandwidth of the gantry stage is lower than that of the planar stage due to resonances.

VI. EXPERIMENT

A. Experiment of Planar Stage

The specifications of the sampling period and the target trajectory are the same as that in the simulations. The frequency responses of the position loop are shown in Fig. 11. The time responses are shown in Fig. 12.

In the conventional method, the tracking error was achieved at a maximum of 1×10^4 counts. The difference in the simulation is caused by the tuning. To get good results at the settling, the feedforward parameters are adjusted deliberately. On the other hand, in the proposed method, the tracking error was achieved in less than 6×10^2 counts. This error seems to be caused by unmodeled disturbance such as cable force, etc. However, the proposed method achieved an improvement by a factor of 17. Moreover, t_{set} and t_{total} were shortened by the proposed method, as shown in Table III(a).

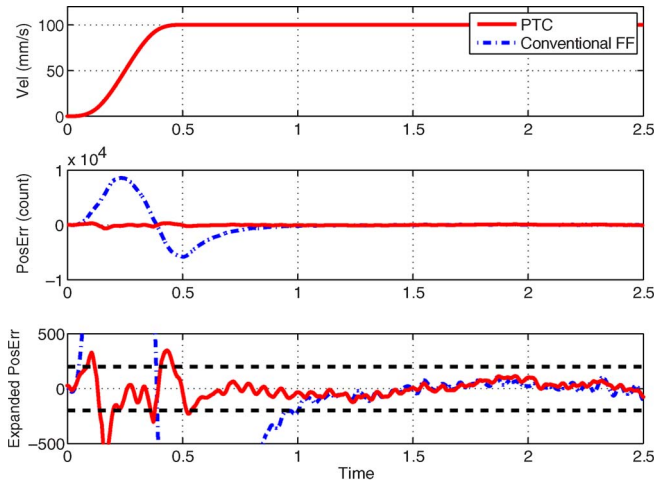


Fig. 12. Experimental results of time response (planar stage).

TABLE III
EXPERIMENTAL RESULTS OF t_{set} AND t_{total} . (a) PLANAR STAGE.
(b) GANTRY STAGE

(a)		
Method	t_{set} [sec]	t_{total} [sec]
Conventional	0.5	1.0
Proposed	0.1	0.6

(b)		
Method	t_{set} [sec]	t_{total} [sec]
Conventional	0.6	1.3
Proposed	0.3	1.0

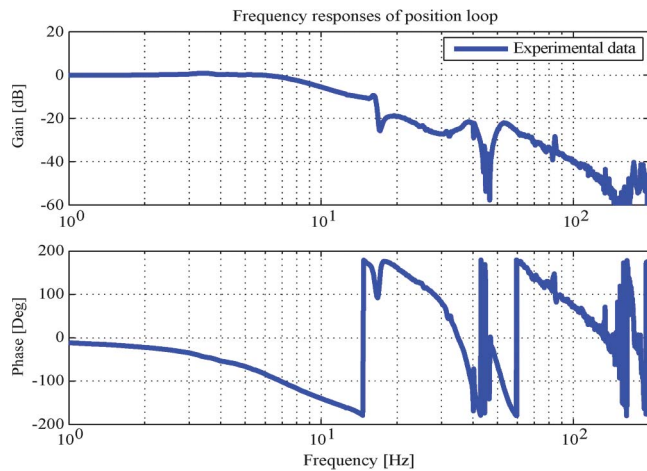


Fig. 13. Frequency responses of a position loop (gantry stage).

B. Experiment of Gantry Stage

The frequency responses of the position loop are shown in Fig. 13. In the same manner as in the simulations, the responses of the gantry stage are worse than those of the planar stage. The time responses are shown in Fig. 14.

The error of the conventional method is below 6×10^4 counts. However, the error of the proposed method is only 1×10^3 counts. Furthermore, t_{set} and t_{total} were improved, as shown in Table III(b).

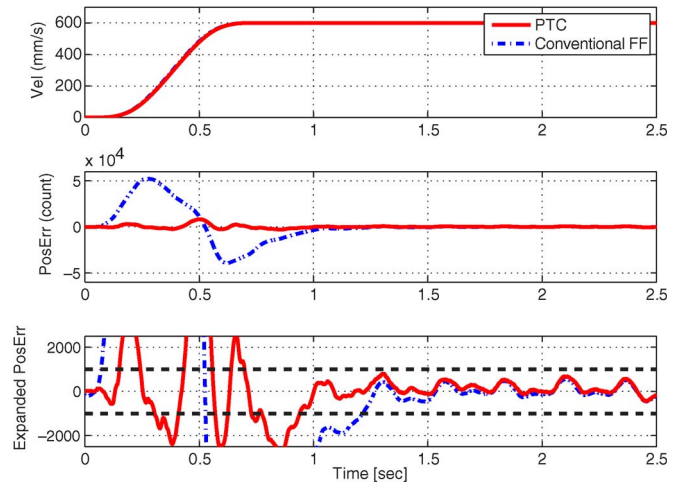


Fig. 14. Experimental results of time response (gantry stage).

VII. CONCLUSION

Since the large-scale high-precision stages have low vibration modes, the high-bandwidth feedback controller cannot be applied. However, the plant dynamics hardly changes because this stage is configured with air guides and linear motors. Therefore, the structure of a 2-DOF control system is generally required, and the high performance needs to be achieved with the help of a feedforward controller. In other words, how well an accurate feedforward controller can be realized is the key to success.

In this paper, for both the proposed and conventional methods, the identical feedback controller was used. Therefore, it is fair to say that the results of the comparison represent the differences in the feedforward methods. In the PTC scheme, the sample holders, which are added to the actual digital controller, are considered and given the discretized plant models. Using this model, the multirate controller is directly realized without the digital redesign. It is possible that an accurate feedforward controller can be realized.

The results of this paper prove that the PTC method is a very effective strategy for achieving high-speed and high-precision tracking control.

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